

EE115A Analog Circuits - Homework 7 Solutions

Due date: 12/11/2025

Note: These solutions are provided for your reference. Please ensure you understand the steps and can reproduce the calculations. As per the instructions, Exercise 5 requires Multisim screenshots which you must generate yourself.

Exercise 1

Given Parameters:

- Technology: $0.18\ \mu\text{m}$
- $k'_n = 4k'_p = 400\ \mu\text{A}/\text{V}^2 \implies k'_p = 100\ \mu\text{A}/\text{V}^2$
- $V_{tn} = -V_{tp} = 0.4\ \text{V}$
- $V_{DD} = 0.9\ \text{V}$, $V_{SS} = -0.9\ \text{V}$
- $I_{REF} = 200\ \mu\text{A}$
- $V_{ov} = 0.2\ \text{V}$ for all transistors.

(a) DC Design

Current Analysis:

- Q_8 and Q_5 form a current mirror. Q_8 and Q_7 form a current mirror.
- $I_{D8} = I_{REF} = 200\ \mu\text{A}$.
- Therefore, $I_{D5} = I_{D7} = 200\ \mu\text{A}$.
- Q_1 and Q_2 are the differential pair, splitting I_{D5} equally (since A and B are grounded).

$$I_{D1} = I_{D2} = I_{D5}/2 = 100\ \mu\text{A}$$

- Q_3 and Q_4 form a current mirror load. $I_{D3} = I_{D1} = 100\ \mu\text{A}$.

$$I_{D4} = I_{D3} = 100\ \mu\text{A} \quad (\text{Matches } I_{D2})$$

- Q_6 is the second stage amplifying transistor. The problem states Q_6 conducts $200\ \mu\text{A}$.

$$I_{D6} = 200\ \mu\text{A}$$

W/L Calculations: Using the saturation current formula:

$$I_D = \frac{1}{2} k' \left(\frac{W}{L} \right) V_{ov}^2 \implies \left(\frac{W}{L} \right) = \frac{2I_D}{k' V_{ov}^2}$$

We calculate the constant factor for NMOS and PMOS:

- For NMOS (Q_1, Q_2, Q_5, Q_7, Q_8):

$$\frac{2}{k'_n V_{ov}^2} = \frac{2}{400 \times (0.2)^2} = \frac{2}{16} = 0.125 (\mu A)^{-1}$$

- For PMOS (Q_3, Q_4, Q_6):

$$\frac{2}{k'_p V_{ov}^2} = \frac{2}{100 \times (0.2)^2} = \frac{2}{4} = 0.5 (\mu A)^{-1}$$

Table 1: Transistor Sizing Results for Exercise 1

Device	Type	$I_D (\mu A)$	$k' (\mu A/V^2)$	Calc ($2I_D/k' V_{ov}^2$)	W/L
Q_1	NMOS	100	400	0.125×100	12.5
Q_2	NMOS	100	400	0.125×100	12.5
Q_3	PMOS	100	100	0.5×100	50
Q_4	PMOS	100	100	0.5×100	50
Q_5	NMOS	200	400	0.125×200	25
Q_6	PMOS	200	100	0.5×200	100
Q_7	NMOS	200	400	0.125×200	25
Q_8	NMOS	200	400	0.125×200	25

DC Output Voltage (V_O): Ideally, if the circuit is perfectly balanced and V_{DS6} and V_{DS7} adjust to satisfy current continuity, the node voltage is typically centered between the rails. Given $V_{DD} = 0.9V$ and $V_{SS} = -0.9V$, $V_O \approx 0V$.

(b) Input Common-Mode Range

The input common-mode range is limited by keeping Q_1/Q_2 in saturation (upper limit) and Q_5 in saturation (lower limit).

Lower Limit ($V_{ICM,min}$): Q_1 and Q_5 must be in saturation. The limiting factor is usually Q_5 entering triode region.

$$V_{in} - V_{GS1} = V_{S1} \implies V_{S1} \geq V_{SS} + V_{ov5}$$

$$V_{in} \geq V_{SS} + V_{ov5} + V_{GS1}$$

Where $V_{GS1} = V_{tn} + V_{ov1} = 0.4 + 0.2 = 0.6V$.

$$V_{ICM,min} = -0.9 + 0.2 + 0.6 = -0.1V$$

Upper Limit ($V_{ICM,max}$): Q_1 must remain in saturation ($V_{DS1} \geq V_{ov1}$).

$$V_{D1} = V_{DD} - V_{SG3} = V_{DD} - (|V_{tp}| + V_{ov3}) = 0.9 - (0.4 + 0.2) = 0.3V$$

For Q_1 saturation: $V_{D1} \geq V_{G1} - V_{tn}$.

$$0.3 \geq V_{ICM} - 0.4 \implies V_{ICM} \leq 0.7V$$

$$V_{ICM,max} = 0.7V$$

Range: $[-0.1V \text{ to } 0.7V]$

(c) Allowable Output Voltage Range

Limited by Q_6 and Q_7 remaining in saturation.

Maximum ($V_{o,max}$): Q_6 stays in saturation: $V_{SD6} \geq V_{ov6}$.

$$V_{DD} - V_o \geq 0.2 \implies V_o \leq 0.9 - 0.2 = \mathbf{0.7V}$$

Minimum ($V_{o,min}$): Q_7 stays in saturation: $V_{DS7} \geq V_{ov7}$.

$$V_o - V_{SS} \geq 0.2 \implies V_o \geq -0.9 + 0.2 = \mathbf{-0.7V}$$

Range: $[-0.7V \text{ to } 0.7V]$

(d) Voltage Gain

$V_A = 6V$. Gain formula: $A_v = A_1 \times A_2$.

Stage 1 parameters (Q_1, Q_2, Q_3, Q_4):

- $I_D = 100 \mu A$.
- $g_{m1} = \frac{2I_D}{V_{ov}} = \frac{200 \mu A}{0.2V} = 1 \text{ mA/V} = 1 \text{ mS}$.
- $r_{o2} = \frac{V_A}{I_D} = \frac{6}{100 \mu A} = 60 \text{ k}\Omega$.
- $r_{o4} = \frac{V_A}{I_D} = 60 \text{ k}\Omega$.
- $R_{out1} = r_{o2} || r_{o4} = 30 \text{ k}\Omega$.
- $A_1 = g_{m1} R_{out1} = 1 \text{ mS} \times 30 \text{ k}\Omega = 30 \text{ V/V}$.

Stage 2 parameters (Q_6, Q_7):

- $I_D = 200 \mu A$.
- $g_{m6} = \frac{2I_D}{V_{ov}} = \frac{400 \mu A}{0.2V} = 2 \text{ mA/V} = 2 \text{ mS}$.
- $r_{o6} = \frac{V_A}{I_D} = \frac{6}{200 \mu A} = 30 \text{ k}\Omega$.
- $r_{o7} = \frac{6}{200 \mu A} = 30 \text{ k}\Omega$.
- $R_{out2} = r_{o6} || r_{o7} = 15 \text{ k}\Omega$.
- $A_2 = -g_{m6} R_{out2} = -2 \text{ mS} \times 15 \text{ k}\Omega = -30 \text{ V/V}$.

Total Gain:

$$A_v = A_1 \times A_2 = 30 \times (-30) = \mathbf{-900 \text{ V/V}}$$

(Magnitude: 59.1 dB)

Exercise 2

Given Parameters:

- Technology: 65-nm
- $k'_n = 540 \mu A/V^2$, $k'_p = 100 \mu A/V^2$.
- $V_{tn} = -V_{tp} = 0.35V$.
- $V_{DD} = 1.2V$, $V_{SS} = 0V$.
- $V_{ov} = 0.15V$.
- $I_{D1-4} = 200 \mu A$, $I_{D6-7} = 400 \mu A$.

(a) DC Design

We determine W/L using $W/L = \frac{2I_D}{k'_n V_{ov}^2}$. $V_{ov}^2 = 0.0225 \text{ V}^2$.

- **For NMOS (Q_1, Q_2, Q_5, Q_7, Q_8):** Denominator factor $k'_n V_{ov}^2 = 540 \times 0.0225 = 12.15 \mu\text{A}$. For $200 \mu\text{A}$: $W/L = 2(200)/12.15 = 32.9$. For $400 \mu\text{A}$: $W/L = 2(400)/12.15 = 65.8$.
- **For PMOS (Q_3, Q_4, Q_6):** Denominator factor $k'_p V_{ov}^2 = 100 \times 0.0225 = 2.25 \mu\text{A}$. For $200 \mu\text{A}$: $W/L = 2(200)/2.25 = 177.8$. For $400 \mu\text{A}$: $W/L = 2(400)/2.25 = 355.6$.

Table 2: Transistor Sizing Results for Exercise 2

Transistor	$I_D (\mu\text{A})$	Calculation	W/L
Q_1, Q_2	200	$400/12.15$	32.9
Q_3, Q_4	200	$400/2.25$	177.8
Q_5	400	$800/12.15$	65.8
Q_6	400	$800/2.25$	355.6
Q_7	400	$800/12.15$	65.8
Q_8	400 (assumed)	$800/12.15$	65.8

(b) Input Common-Mode Range

$$V_{GS,n} = 0.35 + 0.15 = 0.5 \text{ V}. \quad V_{SG,p} = 0.35 + 0.15 = 0.5 \text{ V}.$$

Lower Limit:

$$V_{ICM,min} = V_{SS} + V_{ov5} + V_{GS1} = 0 + 0.15 + 0.5 = \mathbf{0.65 \text{ V}}$$

Upper Limit:

$$V_{D1} = V_{DD} - V_{SG3} = 1.2 - 0.5 = 0.7 \text{ V}$$

$$V_{ICM,max} = V_{D1} + V_{tn} = 0.7 + 0.35 = \mathbf{1.05 \text{ V}}$$

Range: $[0.65 \text{ V to } 1.05 \text{ V}]$

(c) Allowable Output Voltage Range

$$\mathbf{Max:} \quad V_{DD} - V_{ov6} = 1.2 - 0.15 = \mathbf{1.05 \text{ V}}.$$

$$\mathbf{Min:} \quad V_{SS} + V_{ov7} = 0 + 0.15 = \mathbf{0.15 \text{ V}}.$$

Range: $[0.15 \text{ V to } 1.05 \text{ V}]$

(d) Voltage Gain

$$V_A = 2.4 \text{ V}.$$

$$\mathbf{Stage 1} \quad (I_D = 200 \mu\text{A}): \quad g_{m1} = \frac{2(200 \mu\text{A})}{0.15} = 2.67 \text{ mA/V}. \quad r_{o2} = r_{o4} = \frac{2.4}{200 \mu\text{A}} = 12 \text{ k}\Omega. \quad R_{out1} = 6 \text{ k}\Omega.$$

$$A_1 = g_{m1} R_{out1} = 2.67 \text{ m} \times 6 \text{ k} = 16 \text{ V/V}.$$

$$\mathbf{Stage 2} \quad (I_D = 400 \mu\text{A}): \quad g_{m6} = \frac{2(400 \mu\text{A})}{0.15} = 5.33 \text{ mA/V}. \quad r_{o6} = r_{o7} = \frac{2.4}{400 \mu\text{A}} = 6 \text{ k}\Omega. \quad R_{out2} = 3 \text{ k}\Omega.$$

$$A_2 = -g_{m6} R_{out2} = -5.33 \text{ m} \times 3 \text{ k} = -16 \text{ V/V}.$$

$$\mathbf{Total Gain:} \quad A_v = 16 \times (-16) = \mathbf{-256 \text{ V/V}}.$$

Exercise 3

Given:

- 0.13- μm technology ($L_{min} = 0.13 \mu\text{m}$).
- $L_{ov} = 0.1L$.
- $\mu_n = 400 \text{ cm}^2/\text{V} \cdot \text{s}$.
- $V'_A = 5 \text{ V}/\mu\text{m}$.
- $V_{ov} = 0.2 \text{ V}$.

Formulas:

1. Intrinsic Gain (A_0):

$$A_0 = g_m r_o = \left(\frac{2I_D}{V_{ov}} \right) \left(\frac{V_A L}{I_D} \right) = \frac{2V'_A L}{V_{ov}}$$

Using $V'_A = 5 \text{ V}/\mu\text{m}$, $V_{ov} = 0.2 \text{ V}$:

$$A_0 = \frac{2 \times 5 \times L}{0.2} = 50L \text{ (where } L \text{ is in } \mu\text{m)}$$

2. Unity-gain Frequency (f_T):

$$f_T \simeq \frac{3\mu_n V_{ov}}{4\pi L^2}$$

Be careful with units. $\mu_n = 400 \text{ cm}^2/\text{Vs} = 400 \times 10^8 \mu\text{m}^2/\text{Vs}$.

$$f_T = \frac{3 \times (400 \times 10^8) \times 0.2}{4\pi L^2} = \frac{1.91 \times 10^9}{L^2} \text{ Hz} = \frac{1.91}{L^2} \text{ GHz (with } L \text{ in } \mu\text{m)}$$

Table 3: Intrinsic Gain and Unity-Gain Frequency

Length	L (μm)	A_0 (V/V)	f_T (GHz)
L_{min}	0.13	$50 \times 0.13 = \mathbf{6.5}$	$1.91/0.13^2 \approx \mathbf{113}$
$2L_{min}$	0.26	$50 \times 0.26 = \mathbf{13}$	$1.91/0.26^2 \approx \mathbf{28.3}$
$3L_{min}$	0.39	$50 \times 0.39 = \mathbf{19.5}$	$1.91/0.39^2 \approx \mathbf{12.6}$
$4L_{min}$	0.52	$50 \times 0.52 = \mathbf{26}$	$1.91/0.52^2 \approx \mathbf{7.1}$
$5L_{min}$	0.65	$50 \times 0.65 = \mathbf{32.5}$	$1.91/0.65^2 \approx \mathbf{4.5}$

Exercise 4

Given:

- $g_m = 5 \text{ mA/V}$.
- $C_{gs} = 5 \text{ pF}$, $C_{gd} = 1 \text{ pF}$, $C_L = 5 \text{ pF}$.
- $R_{sig} = 10 \text{ k}\Omega$, $R'_L = 10 \text{ k}\Omega$.

Open-Circuit Time Constants (OCTC) Method:

1. Resistance seen by C_{gs} (R_{gs0}):

$$R_{gs0} = R_{sig} = 10 \text{ k}\Omega$$

$$\tau_{gs} = C_{gs}R_{gs0} = 5 \text{ pF} \times 10 \text{ k}\Omega = 50 \text{ ns}$$

2. Resistance seen by C_{gd} (R_{gd0}):

$$R_{gd0} = R_{sig}(1 + g_m R'_L) + R'_L$$

$$g_m R'_L = 5 \text{ mS} \times 10 \text{ k}\Omega = 50$$

$$R_{gd0} = 10 \text{ k}(1 + 50) + 10 \text{ k} = 510 \text{ k} + 10 \text{ k} = 520 \text{ k}\Omega$$

$$\tau_{gd} = C_{gd}R_{gd0} = 1 \text{ pF} \times 520 \text{ k}\Omega = 520 \text{ ns}$$

3. Resistance seen by C_L (R_{L0}):

$$R_{L0} = R'_L = 10 \text{ k}\Omega$$

$$\tau_L = C_L R_{L0} = 5 \text{ pF} \times 10 \text{ k}\Omega = 50 \text{ ns}$$

Results:

• Total Time Constant:

$$\tau_H = \tau_{gs} + \tau_{gd} + \tau_L = 50 + 520 + 50 = \mathbf{620 \text{ ns}}$$

• 3-dB Frequency (f_H):

$$f_H \approx \frac{1}{2\pi\tau_H} = \frac{1}{2\pi(620 \times 10^{-9})} \approx \mathbf{256.7 \text{ kHz}}$$

Percentage caused by interaction of C_{gd} with input: This refers to the Miller effect component in τ_{gd} , specifically the term involving R_{sig} . Term = $C_{gd}R_{sig}(1 + g_m R'_L) = 1 \text{ pF} \times 510 \text{ k}\Omega = 510 \text{ ns}$. Percentage = $\frac{510}{620} \approx \mathbf{82.3\%}$.

To double f_H : We need τ_H to be halved: $\tau_{H,new} = 310 \text{ ns}$. Assume C and R'_L are fixed, we solve for new R_{sig} .

$$\tau_H = C_{gs}R_{sig} + C_{gd}[R_{sig}(1 + 50) + R'_L] + C_L R'_L$$

$$310 \text{ ns} = 5 \text{ p}(R_{sig}) + 1 \text{ p}(51R_{sig} + 10 \text{ k}) + 50 \text{ n}$$

$$310 \text{ n} = 5 \text{ p}R_{sig} + 51 \text{ p}R_{sig} + 10 \text{ n} + 50 \text{ n}$$

$$250 \text{ n} = 56 \text{ p}R_{sig}$$

$$R_{sig} = \frac{250 \times 10^{-9}}{56 \times 10^{-12}} \approx 4464 \Omega \approx \mathbf{4.46 \text{ k}\Omega}$$

Exercise 5

Note: This section provides theoretical calculations. You must set up the circuit in Multisim as shown in Fig P5 to get the simulation results.

Given:

- $g_m = 4 \text{ mA/V}$, $r_o = 20 \text{ k}\Omega$.
- $R_{sig} = 20 \text{ k}\Omega$, $R_L = 20 \text{ k}\Omega$.
- $C_{gs} = 2 \text{ pF}$, $C_{gd} = 0.3 \text{ pF}$, $C_{db} = 0.2 \text{ pF}$, $C_L(\text{total}) = 1 \text{ pF}$.

(a) CS Amplifier

Low-Frequency Gain (A_M):

$$A_M = \frac{v_o}{v_{sig}} = \left(\frac{R_{in}}{R_{in} + R_{sig}} \right) (-g_m R_{out})$$

For MOSFET at low frequency, $R_{in} \rightarrow \infty$, so $\frac{R_{in}}{R_{in} + R_{sig}} \approx 1$.

$$R_{out} = r_o || R_L = 20k || 20k = 10 \text{ k}\Omega$$

$$A_M = -(4 \text{ mS})(10 \text{ k}\Omega) = -\mathbf{40 \text{ V/V}}$$

Estimate f_H using OCTC:

- τ_{gs} : $R_{gs} = R_{sig} = 20 \text{ k}\Omega$. $\tau_{gs} = 2 \text{ pF} \times 20k = 40 \text{ ns}$.
- τ_{gd} : $R_{gd} = R_{sig}(1 + g_m R_{out}) + R_{out} = 20k(1 + 40) + 10k = 820k + 10k = 830 \text{ k}\Omega$. $\tau_{gd} = 0.3 \text{ pF} \times 830k = 249 \text{ ns}$.
- τ_L : $R_{Leq} = R_{out} = 10 \text{ k}\Omega$. $\tau_L = 1 \text{ pF} \times 10k = 10 \text{ ns}$.

$$\tau_H = 40 + 249 + 10 = 299 \text{ ns}$$

$$f_H \approx \frac{1}{2\pi(299\text{ns})} \approx \mathbf{532 \text{ kHz}}$$

Gain-Bandwidth Product (GBW):

$$GBW = |A_M| \times f_H = 40 \times 532 \text{ kHz} = \mathbf{21.28 \text{ MHz}}$$

(b) Cascode Amplifier

A Common-Gate (CG) stage is added. The voltage gain of the first stage becomes small ($A_{v1} \approx -g_m/g_m = -1$), suppressing the Miller effect.

New A_M : The total DC gain of a cascode with $R_L \approx r_o$ is approximately the same as the CS stage because the output resistance is dominated by R_L . $R_{out,cascode} \approx (g_m r_o^2) || R_L$. Since $g_m r_o^2 \approx 1.6 \text{ M}\Omega$ and $R_L = 20 \text{ k}\Omega$, $R_{out} \approx 20 \text{ k}\Omega$.

$$A_M \approx -g_m R_L = -4\text{m} \times 20k = -\mathbf{80 \text{ V/V}}$$

New f_H : Miller effect is removed.

- **Input Node:** $\tau_{in} = R_{sig}(C_{gs1} + C_{gd1}(1 + \frac{g_{m1}}{g_{m2}})) \approx 20k(2\text{p} + 0.3\text{p}(2)) = 20k(2.6\text{p}) = 52 \text{ ns}$.
- **Output Node:** This is usually the dominant pole. $R_{out} \approx R_L = 20 \text{ k}\Omega$. Total capacitance at output is $C_L + C_{gd2} = 1 \text{ pF} + 0.3 \text{ pF} = 1.3 \text{ pF}$. $\tau_{out} = 20k \times 1.3\text{p} = 26 \text{ ns}$.
- **Intermediate Node:** $R \approx 1/g_{m2} = 250\Omega$. Capacitance $\approx C_{db1} + C_{gs2} = 0.2\text{p} + 2\text{p} = 2.2\text{p}$. $\tau = 0.55 \text{ ns}$ (negligible).

$$\tau_H \approx 52 + 26 = 78 \text{ ns}$$

$$f_H \approx \frac{1}{2\pi(78\text{ns})} \approx \mathbf{2.04 \text{ MHz}}$$

New GBW:

$$GBW = 80 \times 2.04 \text{ MHz} \approx \mathbf{163.2 \text{ MHz}}$$